Translational Bioinformatics

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Biomedical Informatics

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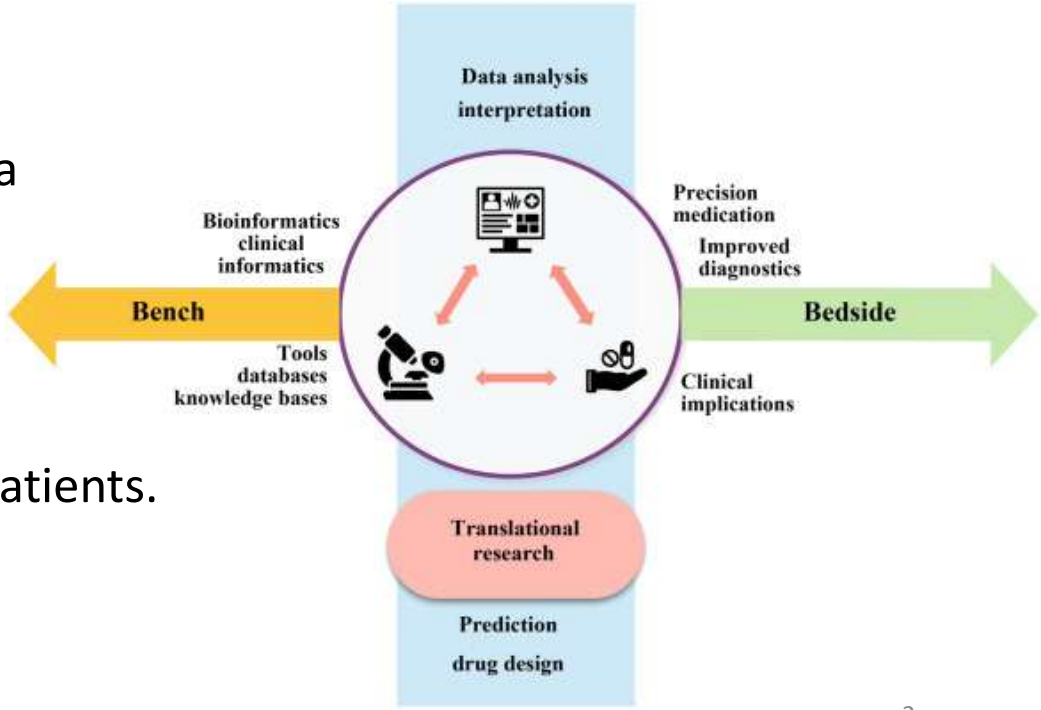
What is translational bioinformatics?

Coined in the early 2010s

Tools and techniques to analyze biological data

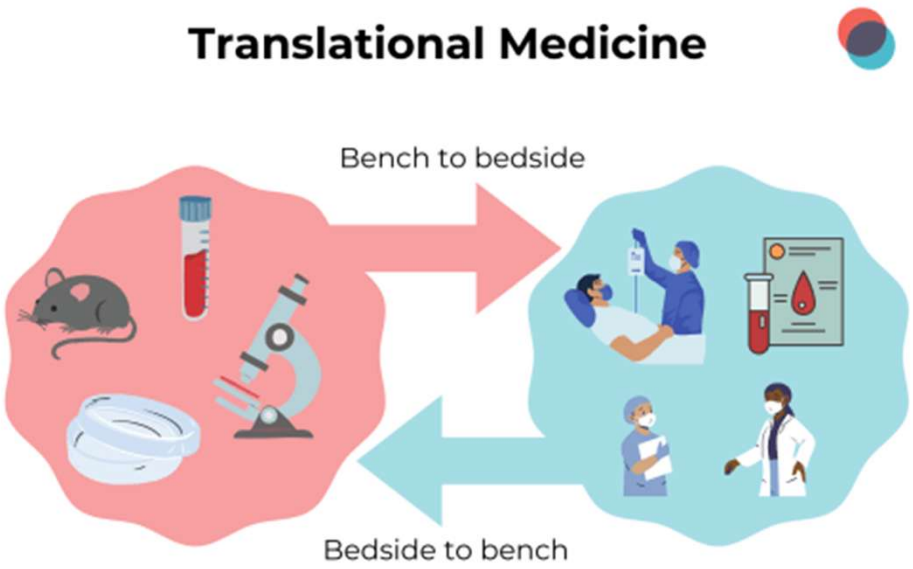
In order to:

- develop new drugs,
- improve disease diagnosis, and
- personalize treatment plans for individual patients.



Translational Medicine

basic science -> clinical research -> application (very slow)



Career path

Medical Student
2009-2015

University of Szeged

Medicine → MD

PhD Student
2015-2019

University of Szeged

Federal University of Minas Gerais

Viral transcriptomics → PhD

Postdoctoral Researcher
2019-present

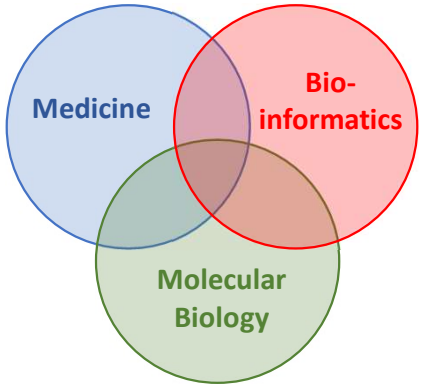
University of Zurich

University Hospital of Zurich

Cell-free DNA-sequencing

Single-cell omics

Translational Bioinformatics



Personally acquired funding

Nemzeti Tehetség Program

UNKP

CAMPUS MUNDI

University of Zurich[UZH] Forschungskredit

Awards

Pro Laudanda Promotione

Stiftung Pfizer Forschungspreis

Other funding

krebsliga

HMZ Stress

Collaborators

USZ Universitäts Spital Zürich

HARVARD MEDICAL SCHOOL

Amsterdam UMC University Medical Centers

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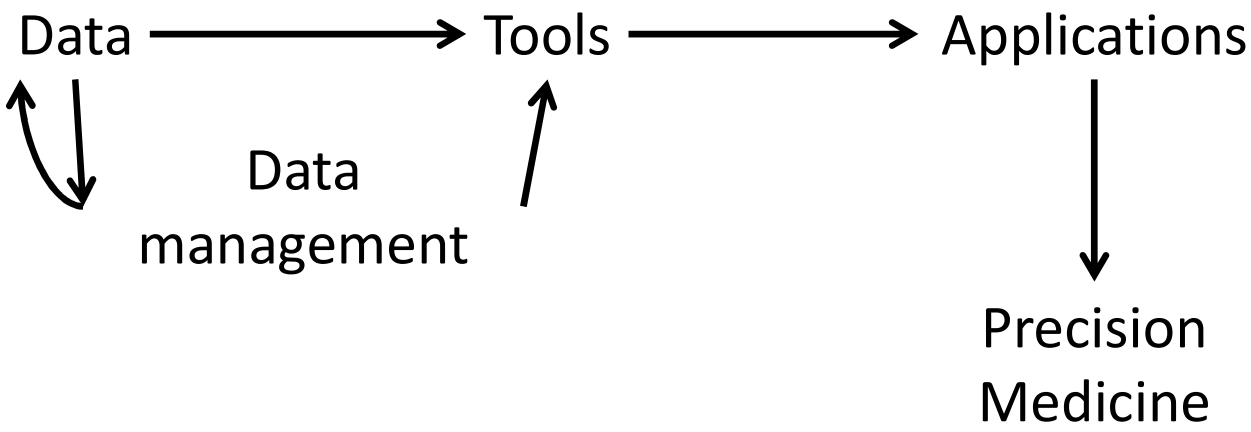
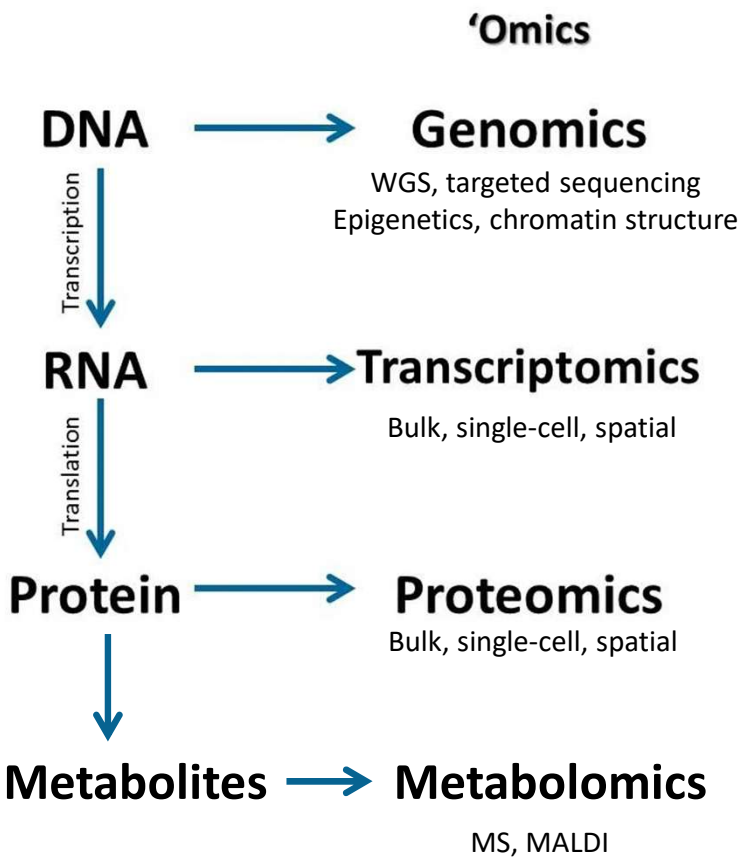


What can we measure?



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Radioimaging → **Radiomics**

Electronic health records

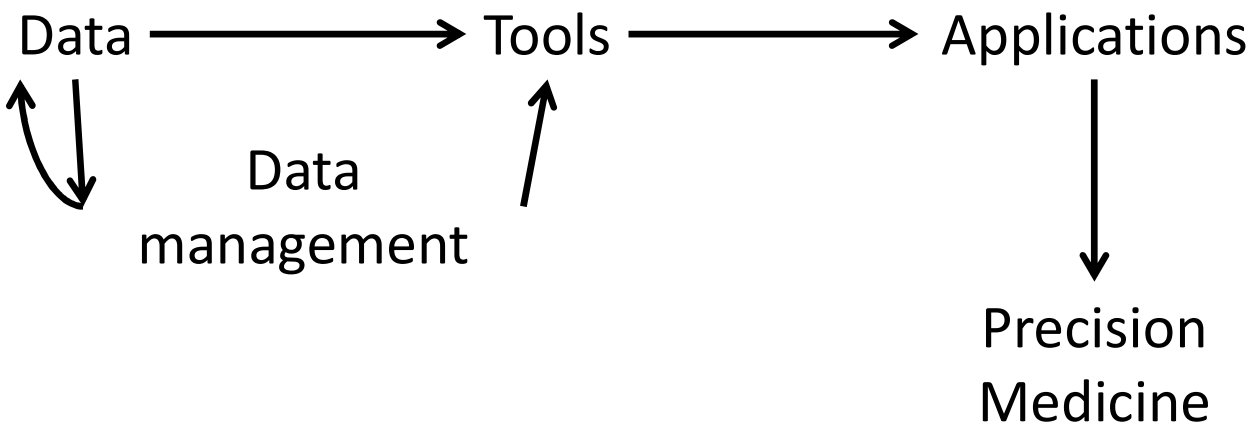
Wearables

Etc.

Characteristics of the data

- Novel technologies $
- Few observations, a great number of parameters
- Multimodal (different technologies combined, on one individual or sample)
- Complex and noisy (heterogenous cohorts, few, often multimorbid patients)
- Privacy concerns!

Translational Bioinformatics



Data management

federated

(no data exchange – analysis parameters optmized across the network)

centralized

(data combined into single database)

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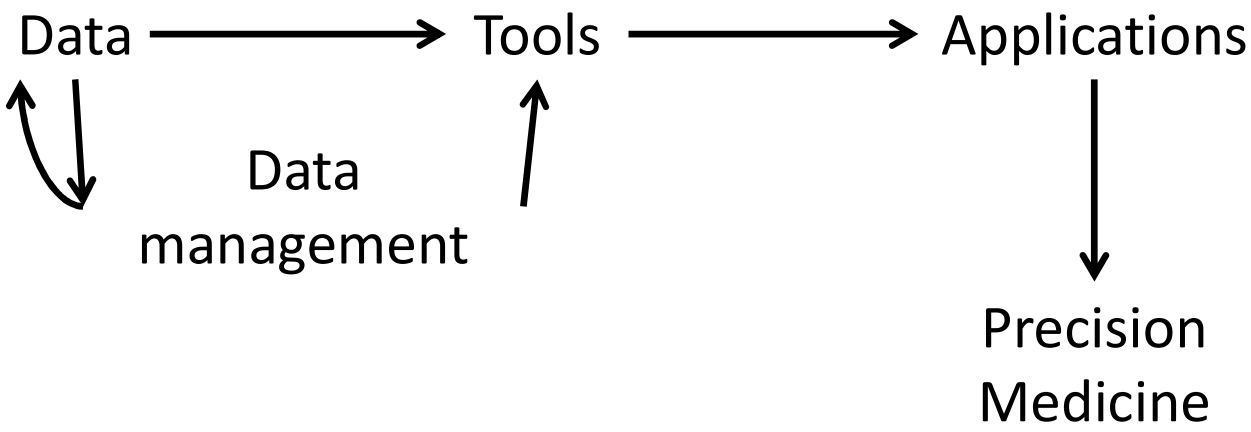


Data analysis in translational bioinformatics

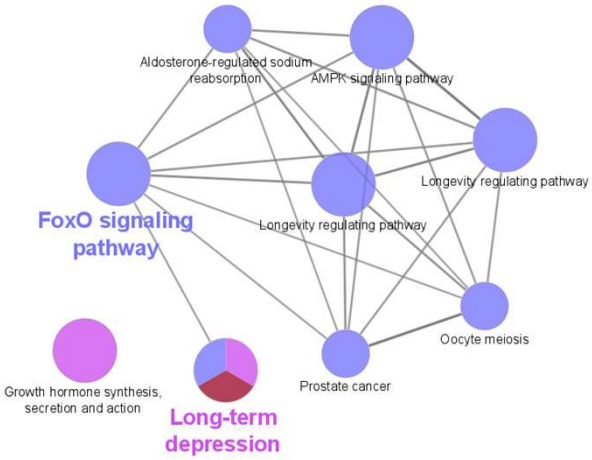
- Highly domain specific
- Traditional statistics
- Machine learning
- Knowledge databases

Big limitation: curse of dimensionality

Overcoming the curse of dimensionality

Examples:

- Highly domain specific → Focusing on genes that could be relevant in the disease
- Traditional statistics → Feature selection methods
- Machine learning → Dimensionality reduction
- Knowledge databases → Pathways and gene sets



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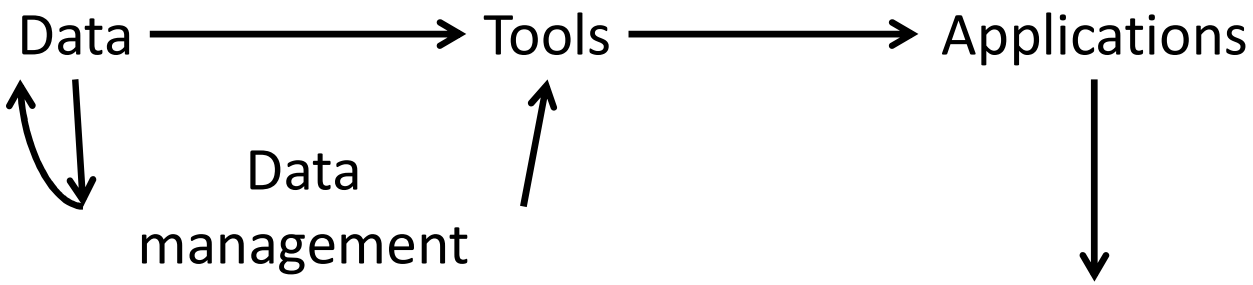
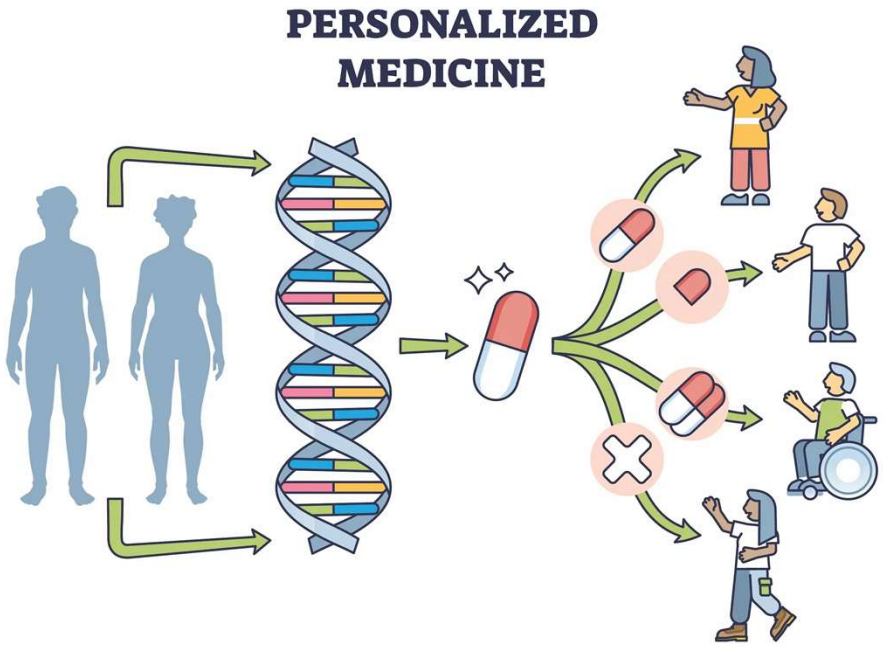


Common applications

Drug Discovery and Development
- Identifying novel targets
- Drug repurposing
- Organoid screening and virtual screening
- Pharmacogenomics

Disease Diagnosis and Prognosis
- Biomarker research
- Prognostic models and classification

Predictive medicine? – Huge trend 10 years ago



Common applications

		Examples
Drug Discovery and Development		
• Identifying novel targets	→	CRISPR-screens
• Drug repurposing	→	Identifying positive side effects from EHRs
• Organoid screening and virtual screening	→	Train models on organoid data to predict *in vivo* response
• Pharmacogenomics	→	Chemotherapy (5-FU) toxicity based on variants
Disease Diagnosis and Prognosis		
• Biomarker research	→	Identify protein in e.g. blood for early diagnosis
• Prognostic models and classification	→	Identify cell-types associated with poor drug response
Predictive medicine	→	Forecast individual disease risks based on genomics

Translational Bioinformatics

Data analytics interface between clinicians and biologists

Driven by

- Cheaper sequencing -> omics
- Machine learning
- More and more data collected

Limitated by

- Ethical considerations
- Privacy issues
- (lacking) Data standards